CONCENTRATION DEPENDENCE OF THE INTERDIFFUSION COEFFICIENT IN BINARY METALLIC SOLID SOLUTION*

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In binary alloys the interdiffusion coefficient $\mathscr{D}_{AB}$ experimentally determined from Fick's second law, is the product of an activity-corrected diffusion coefficient D_{AB} and a thermodynamic factor which represents the departure from ideality of the solution

$$\mathscr{D}_{AB} = D_{AB}\,\alpha_{AB} = D_{AB}\left(1 + \frac{d \ln \gamma_A}{d \ln N_A}\right)$$

Analyses of all the available experimental results concerning solid binary alloys, where solubility is unlimited, leads to the following general empirical correlation

$$D_{AB} = (D^0_{A(B)})^{N_B}(D^0_{B(A)})^{N_A} \exp - \left(\frac{16\Delta T_s^{AB}}{RT}\right)$$

where $D^0_{A(B)}$ and $D^0_{B(A)}$ are the limiting values of the interdiffusion coefficient at infinite dilution and ΔT_s^{AB} represents the difference between the solidus temperature of the phase diagram T_s^{AB} and the temperature which would be computed if the solidus line varied linearly with composition (See Fig. 2).

This correlation is justified by previously successful empirical correlations between diffusivity and physical properties.

RELATION ENTRE LA CONCENTRATION ET LE COEFFICIENT D'INTERDIFFUSION DANS LES SOLUTIONS BINAIRES METALLIQUES

Dans les alliages binaires, le coefficient d'interdiffusion déterminé expérimentalement à partir de la deuxième loi de Fick, est le produit d'un coefficient de diffusion exprimé à partir des activités et d'un facteur thermodynamique représentant l'écart de la solution par rapport à une solution idéale

$$\mathscr{D}_{AB} = D_{AB}\,\alpha_{AB} = D_{AB}\left(1 + \frac{d \ln \gamma_A}{d \ln N_A}\right)$$

Les analyses de tous les résultats expérimentaux valables concernant les alliages solides binaires sans limites de solubilité, conduisent à la relation générale empirique suivante

$$D_{AB} = (D^0_{A(B)})^{N_B}\,(D^0_{B(A)})^{N_A} \exp - \left(\frac{(16\,\Delta T_s^{AB}}{RT}\right)$$

où $D^0_{A(B)}$ et $D^0_{B(A)}$ sont les valeurs limites du coefficient d'interdiffusion pour une dilution infinie et ΔT_s^{AB} représente la différence entre la température du solidus du diagramme de phase T_s^{AB} et la température qui serait évaluée si la ligne du solidus variait linéairement avec la composition (voir Fig. 2).

Cette relation est justifiée par des relations empiriques entre la diffusivité et les propriétés physiques qui se sont révélées jusqu'à maintenant très satisfaisantes.

KONZENTRATIONSABHÄNGIGKEIT DES DIFFUSIONSKOEFFIZIENTEN IN BINÄREN, METALLISCHEN MISCHKRISTALLEN

Der experimentell aus dem 2. Fick'schen Gesetz bestimmte Diffusionskoeffizient $\mathscr{D}_{AB}$ in binären Legierungen ist das Produkt aus einem aktivitäts-korrigierten Diffusionskoeffizienten D_{AB} und einem thermodynamischen Faktor, der die Abweichung von der idealen Lösung beschreibt

$$\mathscr{D}_{AB} = D_{AB}\,\alpha_{AB} = D_{AB}\left(1 + \frac{d \ln \gamma_A}{d \ln N_A}\right).$$

Die Analyse der verfügbaren experimentellen Ergebnisse an binären, festen Legierungen mit unbegrenzter Löslichkeit führt zur folgenden allgemeinen empirischen Beziehung

$$D_{AB} = (D^0_{A(B)})^{N_B}\,(D^0_{B(A)})^{N_A} \exp - \left(\frac{16\Delta T_s^{AB}}{RT}\right).$$

Dabei sind $D^0_{A(B)}$ und $D^0_{B(A)}$ die Grenzwerte der Inter-Diffusionskoeffizienten bei unendlicher Verdünnung und ΔT_s^{AB} steht für die Differenz zwischen der Solidustemperatur des Phasendiagramms T_s^{AB} und der Temperatur, die man errechnen würde, wenn sich die Soliduslinie linear mit der Zusammensetzung änderte (s. Fig. 2).

Diese Beziehung ist durch frühere erfolgreiche empirische Zusammenhänge zwischen Diffusionsvermögen und physikalischen Eigenschaften gerechtfertigt.

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In binary mixtures where solubility is unlimited the various diffusion coefficients, which are directly measurable by experiment are illustrated schematically in Fig. 1. $\mathscr{D}_{AB}$ is the interdiffusion coefficient, D_A^* and D_B^* are the self-diffusion coefficients of species A and B in the mixture. The limiting values of these diffusion coefficients are given on Fig. 1.

A fairly convincing argument has been proposed by Darken[1] for the use of two intrinsic diffusion coefficients, in binary systems exhibiting substitutional diffusion. The Darken theory as re-examined recently by Manning[2] led to the well-known relation

$$\mathscr{D}_{AB} = (x_B D_A^* + x_A D_B^*)\alpha_{AB} S_{AB} \qquad (1)$$

where α_{AB} is the thermodynamic factor, S_{AB} is a factor introduced by Manning[2] which arises from the vacancy flow effect.

So, in the last years, the current trend has been to consider the interdiffusion coefficient as a composite coefficient as shown by equation (1).

When sufficient data of this type are obtained this approach may be the more correct from a fundamental point of view, but the amount of such information now available is limited to a few systems.

Fortunately, the Matano interface has been found to remain at the initial interface within experimental error so that one diffusion coefficient (which may depend upon concentration) adequately describes the distribution of both components of the system after diffusion. Furthermore, in phase transformation problems, the diffusion coefficients which are needed to compute the rates of phase growth are the interdiffusion coefficients in each phase.

So a study of the nature of the concentration dependence of the interdiffusion coefficient may be of great help.

This concentration dependence has been demonstrated for many pairs of metals. However, the nature of this dependence has never been fully elucidated. As no correlation can be deduced from a theoretical point of view, the currently popular approach of examining diffusion data in the light of previously successful empirical correlations between diffusivity and certain physical properties will be used in this study. Many investigators have suspected that the concentration dependence of the interdiffusion coefficient could be related to the thermodynamic properties of the solutions. The concentration dependence of this coefficient is certainly related to the thermodynamic properties of the solution because the true driving force of diffusion is the chemical potential gradient of each component, and from the phenomenological theory of diffusion, based on thermodynamics or irreversible processes,[3] it can be be shown that the interdiffusion coefficient experimentally determined $\mathscr{D}_{AB}$ is the product of an activity-corrected diffusion coefficient, D_{AB}, and a thermodynamic factor

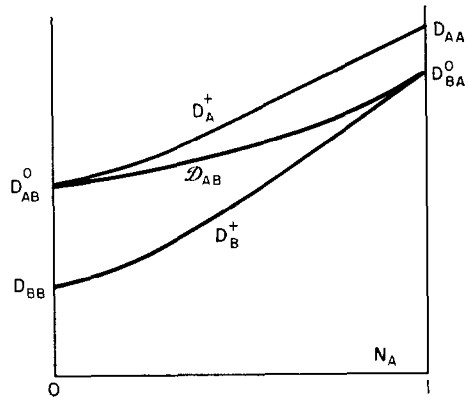

Fig. 1. Schematic representation of the various diffusion coefficients in a binary alloy.

$$\mathscr{D}_{AB} = D_{AB}\,\alpha_{AB} \qquad (2)$$

where

$$\alpha_{AB} = 1 + \frac{d \ln \gamma_A}{d \ln N_A} = 1 + \frac{d \ln \gamma_B}{d \ln N_B}$$

where γ_A is the activity coefficient and N_A the atomic fraction of component A.

A preceding analysis for liquid binary solutions[4] has led one of the authors to the following general result: the activity-corrected interdiffusion coefficient D_{AB} varies exponentially with the mole fraction of one of the components of the solution, between the two limiting values of the diffusion coefficient (see Fig. 1)

$$D_{AB} = (D_{AB}{}^0)^{N_B}\,(D_{BA}{}^0)^{N_A} \qquad (3)$$

In spite of the fact that this result seemed also valid for diffusion in solid binary alloys, a careful analysis of all the available experimental results has shown that this correlation was not valid for all systems. A new empirical correlation which fits better all available data for systems where solubility is unlimited will be presented below. This correlation is deduced from a qualitative observation of Birchenall[5] and Le Claire.[6] These authors noted that there seems to be a rough correspondence between the melting point of the alloy and the interdiffusion coefficient.

"An increase in the interdiffusion coefficient is attended by a decrease in the melting point of the alloy and a decrease in the interdiffusion coefficient is attended by an increase of the melting point."

The correlation proposed takes into account quantitatively the correspondence between the solidus curve of the phase diagram and the concentration dependence of the interdiffusion coefficient and is an extension of the correlation (equation 3) proposed for liquid solutions.

As it will be shown below, all the available experimental results are satisfactorily represented by the relation

$$\mathscr{D}_{AB} = \alpha_{AB} \cdot D_{AB} = \alpha_{AB}(D_{AB}{}^0)^{N_B}(D_{BA}{}^0)^{N_A} \exp\left(-\frac{16\Delta T_s{}^{AB}}{RT}\right) \quad (4)$$

where α_{AB} is the thermodynamic factor. $D_{AB}{}^0$ and $D_{BA}{}^0$ are the limiting values of the interdiffusion coefficient and $\Delta T_s{}^{AB}$ is defined from the solidus line of the phase diagram and represents the departure from linearity of the solidus line (see Fig. 2)

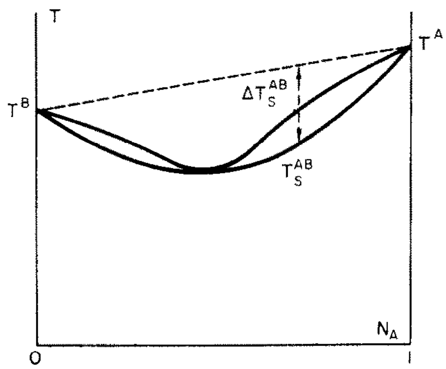

FIG. 2. Schematic representation of the phase diagram of a binary alloy and representation of $\Delta T_s{}^{AB}$ as defined by equation (5).

$$T_s{}^{AB} = N_A T_m{}^A + N_B T_m{}^B + \Delta T_s{}^{AB} \quad (5)$$

$\Delta T_s{}^{AB}$ is the difference between the solidus temperature $T_s{}^{AB}$ and the temperature which would be computed if the solidus line varied linearly with composition.

The factor 16 before $\Delta T_s{}^{AB}$ has been found by trial and error.

For comparison with experimental data we have plotted for each system the concentration dependence of the following coefficients:

The interdiffusion coefficient

$$\mathscr{D}_{AB} = f(N_A) \quad (6)$$

The activity-corrected interdiffusion coefficient $D_{AB} = \mathscr{D}_{AB}/\alpha_{AB}$ computed from $\mathscr{D}_{AB}$ and available thermodynamic data, and

$$\tilde{D}_{AB} = [(\mathscr{D}_{AB}/\alpha_{AB}] \exp\left(\frac{16\,\Delta T_s{}^{AB}}{RT}\right) \quad (7)$$

computed from solidus line.

If the correlation[4] is valid, this last coefficient $\tilde{D}_{AB}$ should vary exponentially with the atomic

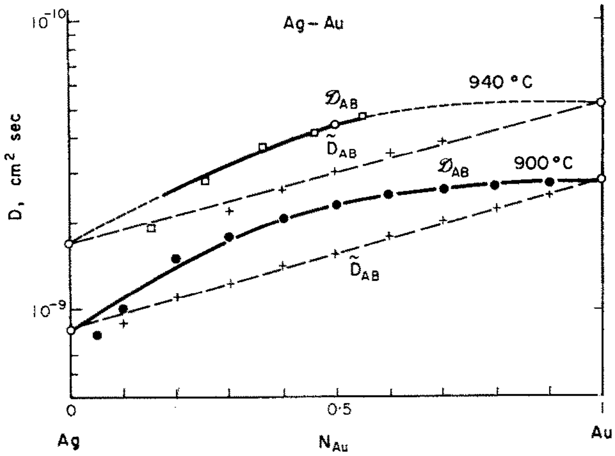

FIG. 3. Concentration dependence of the interdiffusion coefficient in Ag–Au system
● Seith and Kottman[9]
□ Balluffi and Seigle[10]
○ Johnson[11]
⊗ Mallard *et al.*[12]

composition of the system. This relation has been tested on six binary systems: Cu–Ni, Fe–Ni, Cu–Au, Cu–Pd, Ag–Au, Au–Ni, and, as shown on Figs. 3–10, all the data satisfy the relation (4), log $\tilde{D}_{AB}$ computed by equation (7) varies linearly with composition.

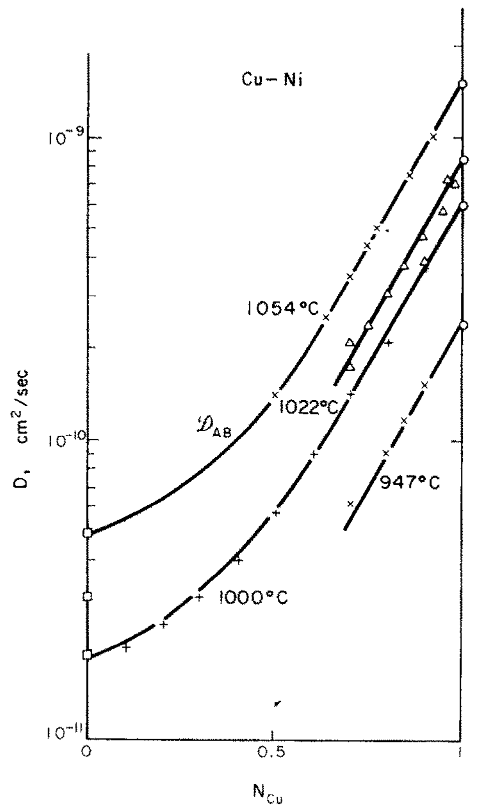

FIG. 4. Concentration dependence of the interdiffusion coefficient in Cu–Ni system
× Da Silva and Mehl[16]
△ Thomas and Birchenall[17]
+ Levasseur and Philibert[18]
⊗ Mackliett[19]
⊠ Muraka et al[20]

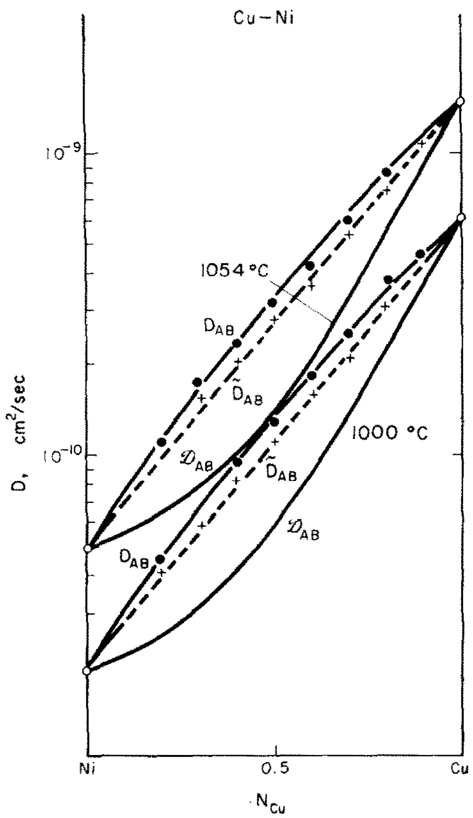

FIG. 5. Concentration dependence of the interdiffusion coefficient in Cu–Ni system and test of equation (4).

DISCUSSION

To justify the correlation proposed, equation (4), assume that the variation with temperature of the activity-corrected interdiffusion coefficient D_{AB} can be described by the Arrhenius law:

$$D_{AB} = D_{0(AB)} \exp\left(-\frac{Q_{AB}}{RT}\right) \qquad (8)$$

where the frequency factor $D_{0(AB)}$ and the activation energy Q_{AB} are concentration dependent.

From the correlation proposed (4), the activation energy Q_{AB} can be expressed as a function of the activation energies of the diffusion coefficients of A into B and of B into A at infinite dilution

$$Q_{AB} = N_B Q^0_{A(B)} + N_A Q^0_{B(A)} + 16\Delta T_s^{AB} \qquad (9)$$

The activation energy for diffusion of B into A at infinite dilution $Q^0_{B(A)}$ can be written as the sum of the activation energy for self-diffusion of A, and an interaction energy Δq_B:

$$Q^0_{B(A)} = Q_{AA} + \Delta q_B \qquad (10)$$

and

$$Q^0_{A(B)} = Q_{BB} + \Delta q_A$$

Using these two expressions, equation (9) can be rewritten

$$Q_{AB} = N_A Q_{AA} + N_B Q_{BB} + 16\Delta T_s^{AB} + N_B \times \Delta q_A + N_A \Delta q_B \qquad (11)$$

It has been shown by Le Claire[6] and Toth and Searcy[7] that the activation energy for self diffusion in a pure metal can be correlated by

$$Q_{AA} = \Delta H_{fV}{}^A + \Delta H_{mV}{}^A = 0{\cdot}27 L_s{}^A + 16 T_m{}^A \qquad (12)$$

where $L_s{}^A$ is the enthalpy of sublimation and $T_m{}^A$ is the melting point of the pure metal.

Replacing Q_{AA} in equation 11 by expression (12) gives, taking into account equation 5,

$$Q_{AB} = 0.27(N_A L_s{}^A + N_B L_s{}^B) + 16T_s{}^{AB} + N_B \Delta q_A + N_A \Delta q_B \qquad (13)$$

Now, in the Bragg–Williams approximation, following

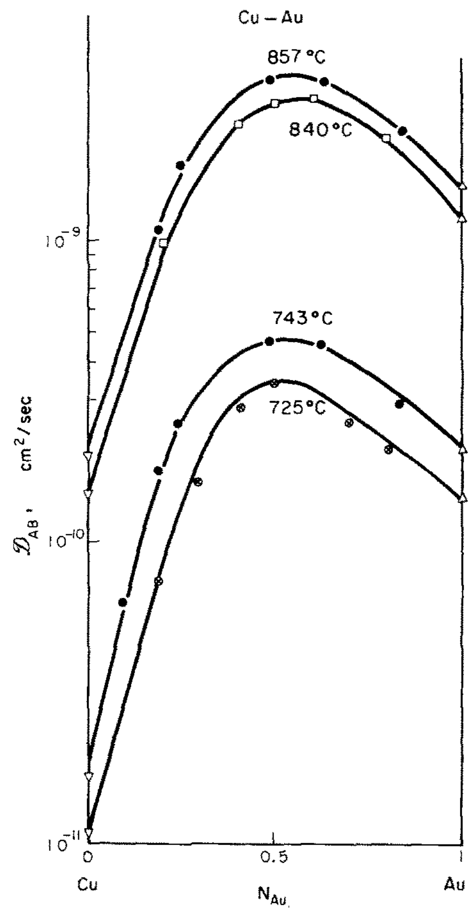

FIG. 6. Concentration dependence of the interdiffusion coefficient in Cu–Au system
⊗ Ziebold[21]
□ Birchenall[5]
● Present study
▽ Le Claire[22]
△ Vignes and Haeussler[23]

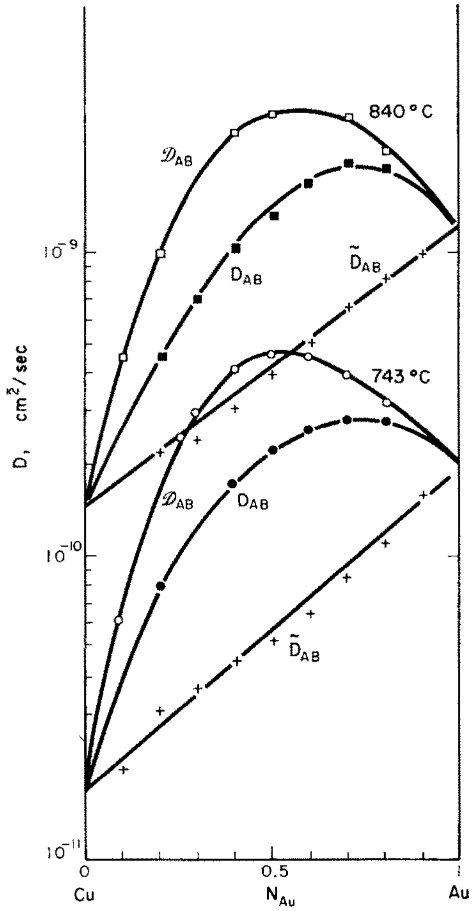

FIG. 7. Concentration dependence of the interdiffusion coefficient in Cu–Au system and test of equation (4).

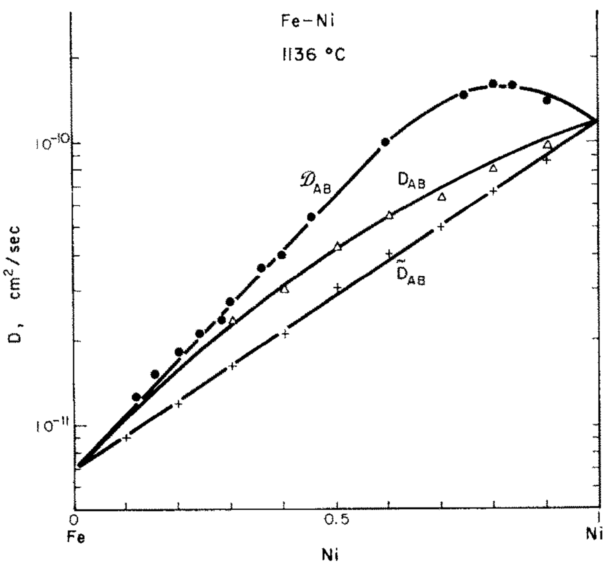

FIG. 8. Concentration dependence of the interdiffusion coefficient in the Fe–Ni system and test of equation (4). ● Badia and Vignes[25]

Toth and Searcy[7] and Shapink[8], the activation energy for vacancy formation in a binary alloy is given by:

$$\Delta H_{fV}{}^{AB} = 0.27(N_A L_s{}^A + N_B L_s{}^B - \Delta H_{\text{mix}}) \quad (14)$$

where ΔH_{mix} is the enthalpy of mixing.

If we assume that the activation energy for vacancy migration in a binary alloy, is by generalization of equation (12), given by

$$\Delta H_{mV}{}^{AB} = 16 T_s{}^{AB} \quad (15)$$

where $T_s{}^{AB}$ is the solidus temperature, the activation energy Q_{AB} (defined by equation 8) can be written as:

$$Q_{AB} = (\Delta H_{fV}{}^{AB} + \Delta H_{mV}{}^{AB}) + N_B \, \Delta q_A + N_A \, \Delta q_B - 0.27 \Delta H_{\text{mix}} \quad (16)$$

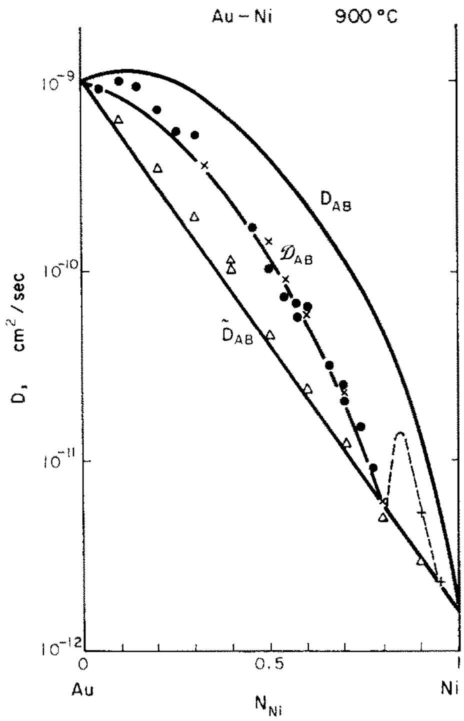

FIG. 9. Concentration dependence of the interdiffusion coefficient in the Au–Ni system and test of equation (4).
● Reynolds *et al.*[27]
× Lifshin[28]

The contribution due to the heat of mixing 0.27 ΔH_{mix} is generally very small and in a first approximation can be neglected.

So, the correlation proposed (equation 4), expresses only the fact that the main contribution to the activation energy for the activity-corrected interdiffusion coefficient is the sum of the activation energies for vacancy formation and migration in a binary alloy. Of course, this derivation is not a demonstration, but

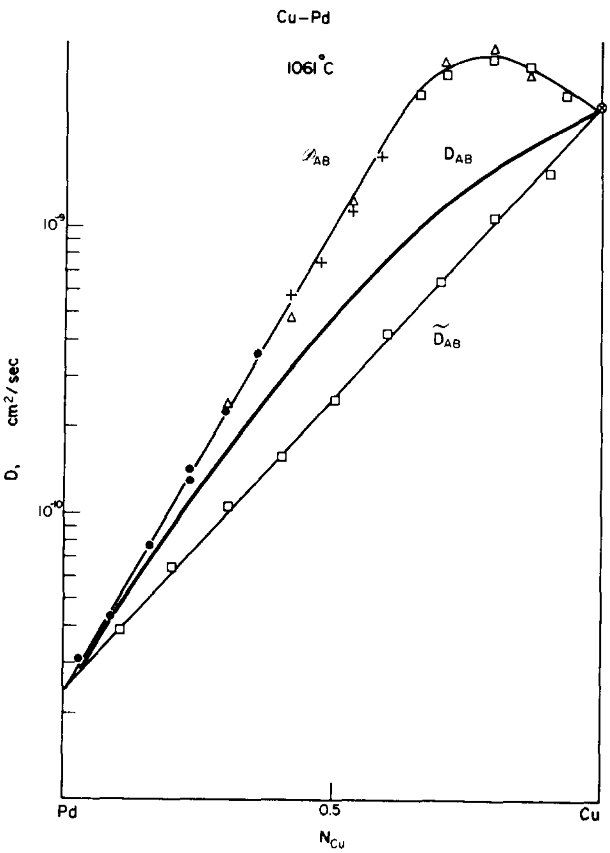

FIG. 10. Concentration dependence of the interdiffusion coefficient in the Cu–Pd system and test of equation (4).
●+□△ Present study
⊗ Peterson[29]

only a justification, and the correlation proposed is a purely empirical one which can be used for predicting the concentration dependence of the interdiffusion coefficient, binary alloys when solubility is unlimited.

ANALYSIS OF EXPERIMENTAL RESULTS

The concentration dependence of the interdiffusion coefficients has been studied, from several diffusion data, for Cu–Ni, Au–Ni, Ag–Au, Fe–Ni systems.

As parts of complete studies of diffusion in Cu–Au, Cu–Pd systems, the concentration dependence of the mutual diffusion coefficient was determined at several temperatures using in each case incremental couples.

As it has been shown by several authors that the pure metal couples yield often higher interdiffusion coefficients than did the incremental couples and that these high values decreased with an increase in the diffusion time, the most reliable values are obtained from the incremental couples.

The concentration gradients were determined with a CAMECA Electron Microprobe, and the diffusion coefficients were calculated by the Matano analysis and, at the two ends of the diffusion couple, by the Hall analysis. Discussion of the results obtained for these two systems will be given below.

Gold–Silver

The concentration dependence of the interdiffusion coefficient has been determined by Seith and Kottman[9] at 900°C, by Bulluffi and Siegle[10] at 940°C. The values of $\mathscr{D}_{AB}$ at $N_{Au} = 0.5$ computed from Johnson's data[11] at 900°C and 940°C are in excellent agreement with those of Refs. 9, 10. These results have been gathered and plotted on Fig. 3 together with the limiting values of $\mathscr{D}_{AB}$ determined by radioactive tracers by Mallard *et al.*[12]. The diffusion isotherm shows a positive departure from linearity.

The thermodynamic factor values have been computed from the relationship given by Hultgren *et al.*[13] for ΔG^{xs}

$$\Delta G^{xs} = -(4850 - 800N_{Ag} - 1{,}375T)_A N_g(1 - N_{Ag})$$

Temp.	N_{Ag}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
900°C	α_{AB}	1.145	1.29	1.421	1.548	1.60	1.628	1.59	1.482	1.289
950°C	α_{AB}	1.00	1.14	1.40	1.58	1.69	1.66	1.60	1.47	1.10

which yields for α_{AB} at 900°C the tabulated values. The values of α_{AB} at 950°C have been computed by Meyer and Slifkin.[14]

For this system, the solidus curve of the phase diagram[15] shows a very slight positive departure from linearity. At 900°C, the following values of $\exp(-16\Delta T_s/2T)$ have been computed.

N_{Au}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
ΔT_s	8	13	16	18	19	18.5	17	12	6
$\exp\left(-\frac{16\,\Delta T_s}{2T}\right)$	0.94	0.915	0.895	0.885	0.88	0.88	0.89	0.925	0.96

Log $\tilde{D}_{AB}$ as defined by equation (7) varies linearly with composition, as shown on Fig. 3, and as predicted by equation (4).

Copper–nickel

The concentration dependence of the interdiffusion coefficient has been determined by several authors, Da Silva and Mehl at 1054 and 947°C.[16] Thomas and Birchenall at 1022°C[17] and Levasseur and Philibert at 1000°C.[18] These results have been gathered and plotted in Fig. 4 together with the limiting values determined by radioactive tracer techniques by Mackliett[19] and Murarka.[20] A very good agreement can be observed between all these results.

In order to compute the activity corrected diffusion coefficient D_{AB}, the thermodynamic factor values were computed graphically on the basis of the activity data reported by Hultgren.[13] The following values of α_{AB} have been obtained at 1054°C and 947°C.

Temp.	N_{Ni}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
1054°C	α_{AB}	0.8	0.65	0.54	0.53	0.45	0.40	0.44	0.58	0.72
947°C	α_{AB}		0.658	0.55	0.51	0.43	0.39	0.42	0.56	

The computed values of the activity corrected diffusion coefficient D_{AB} are plotted on Fig. 5 at 1000°C and 1054°C together with the values of $\mathscr{D}_{AB}$. As can be seen on Fig. 5, log D_{AB} shows a slight positive departure from linearity. The solidus curve of the Cu–Ni phase diagram[15] shows a slight negative departure from linearity. At 1054° the following values of exp $(-16\Delta T_s/2T)$ have been computed.

N_{Ni}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$-\Delta T_s$	10	20	30	25	25	20	15	10	0
$\exp\left(-\frac{16\,\Delta T_s}{2T}\right)$	1.06	1.13	1.20	1.16	1.16	1.13	1.09	1.06	1

As can be seen on Fig. 5 log $\tilde{D}_{AB}$ as defined by equation 7 varies linearly with composition, as predicted by equation (4).

Copper–gold

The concentration dependence of the interdiffusion coefficient has been determined by Birchenall[5] at 840°C, Ziebold[21] at 725°C and by ourselves at 743°C and 840°C. These results have been gathered and plotted on Fig. 6 together with the limiting values as reported by Le Claire[22] and Vignes and Haeussler.[23] Very good agreement can be observed between these results. Log $\mathscr{D}_{AB}$ shows a strong positive departure from linearity and the solidus curve shows a strong negative departure from linearity.

The thermodynamic factor values were computed graphically on the basis of the activity data reported by Hultgren.[13] The following values have been obtained at 840°C and 743°C.

Temp.	N_{Cu}	0.2	0.3	0.4	0.5	0.6	0.7	0.8
870°C	α_{AB}	1.145	1.42	1.78	2.15	2.37	2.6	1.15
743°C	α_{AB}	1.22	1.5	1.8	2.1	2.35	2.6	2.2

As can be seen on Fig. 7, log D_{AB} shows still a strong positive departure from linearity. But the solidus curve of the Cu–Au phase diagram[15] shows a strong negative departure from linearity. The following values of exp $(-16\Delta T_s/2T)$ have been computed.

Temp.	N_{Au}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
	ΔT_s	−50	−100	−150	−170	−180	−175	−155	−115	−60
840°C	$\exp\left(-\frac{16\,\Delta T_s}{2T}\right)$	1.43	2.06	2.95	3.4	3.6	3.5	3	2.28	1.54
743°C	$\exp\left(-\frac{16\,\Delta T_s}{2T}\right)$	1.48	2.2	3.28	3.8	4.17	4	3.3	3.5	1.6

As shown in Fig. 7 $\log \tilde{D}_{AB}$ as defined by equation 7 varies linearly with atomic composition as predicted by equation (4).

Iron–nickel

The concentration dependence of the interdiffusion coefficient has been determined by several authors at various temperatures; the most recent studies are those of Goldstein and Ogilvie,[24] and Badia and Vignes.[25] A good agreement has been observed between the two sets of results. Log $\mathscr{D}_{AB}$ shows a strong positive departure from linearity (Fig. 8). The thermodynamic factor was computed graphically from the activity data of Fleisher and Elliott[26] at 1100°C. The following values have been obtained at 1100°C.

N_{Ni}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
γ_{Ni}	0.34	0.366	0.366	0.4	0.42	0.466	0.56	0.685	0.91
α_{AB}	1	1	1.14	1.276	1.46	1.8	2.3	2.0	1.5

As can be seen on Fig. 8, log D_{AB} still shows a positive deviation from linearity. But the solidus curve of the Fe–Ni diagram[15] shows a negative departure from linearity.

The following values of $\exp(-16\Delta T_s/2T)$ at 1136°C have been computed.

N_{Ni}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$-\Delta T_s$	40	52	60	60	60	50	42	35	20
$\exp\left(-\frac{16\,\Delta T_s}{2T}\right)$	1.26	1.34	1.41	1.41	1.41	1.33	1.27	1.22	1.12

As shown in Fig. 8 log $\tilde{D}_{AB}$ as defined by equation 7 varies linearly with atomic composition as predicted by equation (4).

Gold–nickel

The concentration dependence of the interdiffusion coefficient has been determined by Reynolds, Averbach and Cohen[27] at 875°C, 900°C, 925°C, and by Lifshin[28] at 875°C, 900°C, 925°C.

These two sets of results agree on the gold-rich side of the composition but the values obtained by Reynolds are much too high on the Nickel-rich side. Furthermore as the values of the interdiffusion coefficients reported by Reynolds were obtained from a variety of incremental couples, a set of values, selected from their data and from data of Lifshin in order to draw a "coherent" curve (see Fig. 9) of the diffusion isotherm is given in table 1.

The thermodynamic factor's values (see below) used are those reported by Reynolds[28]

As can be seen log D_{AB} shows a strong positive deviation from ideality.

TABLE 1. Diffusion in Gold–Nickel alloys. Values of D_{AB} used to draw the curve of Fig. 8 from Reynolds *et al.*[28]

x_{Ni}	D_{AB}(cm²/sec) $\times 10^{-11}$
0.05	88
0.1	82
0.15	75
0.20	70
0.25	60
0.25	57.5
0.46	17
0.50	10.3
0.50	14
0.54	7.4
0.55	8.8
0.57	6.74
0.58	5.8
0.60	5.8
0.60	6.28
0.62	4.4
0.65	3.7
0.66	3.2
0.70	2.1
0.70	2.3
0.74	1.54
0.78	0.903
0.1	80
0.2	65
0.25	50
0.34	35
0.6	6
0.8	6.5
0.9	0.63
0.95	0.295

$D^0_{Ni(Au)} = 10^{-9}$ cm²/sec

$D^0_{Au(Ni)} = 1.2 \times 10^{-12}$

N_{Ni}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
α_{AB}	0.72	0.57	0.45	0.38	0.33	0.25	0.16	0.11	0.33

The solidus curve of the Au–Ni diagram[15] shows a very strong negative departure from linearity. At 900°C the following values of exp $(-16\Delta T_s/2T)$ have been computed.

N_{Ni}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$-\Delta T_s$	85	160	220	260	300	325	340	320	240
$\exp\left(-\frac{16\,\Delta T_s}{2T}\right)$	1.79	3	4.46	5.9	7.7	9	10	8.9	5.15

As can be seen on Fig. 9 log $\tilde{D}_{AB}$ as defined by equation 7 varies linearly with composition as predicted by equation (4). (Only the reversal of the diffusion isotherm $\mathscr{D}_{AB}$ near the high nickel and cannot be explained.)

Copper–palladium

The concentration dependence of the interdiffusion coefficient has been determined by Thomas and Birchenall[17] at 878°C, 972°C and 1038°C, but only values of $\mathscr{D}_{AB}$ between 30 and 70% of copper have been reported. These authors have assumed that $\mathscr{D}_{AB}$ was constant in both dilute solutions. As part of a complete study of diffusion in this system, we have determined the variation of the interdiffusion coefficient $\mathscr{D}_{AB}$ with composition, using incremental couples at 1061°C.

As can be seen, on Fig. 10, log $\mathscr{D}_{AB}$ shows a strong positive deviation from linearity. At 1061°C the thermodynamic factor values were computed from activity data reported by Hultgren[13].

N_{Pd}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
α_{AB}	1.85	2.44	2.8	2.6	2.2	1.6	1.37	~1.1	1

At 1061°C the following values of exp $(-16\Delta T_s/2T)$ have been computed.

N_{Pd}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$-\Delta T_s$	43	80	100	110	110	100	90	70	40
$\exp\left(-\frac{16\Delta T_s}{2T}\right)$	1.29	1.62	1.82	1.93	1.93	1.82	1.72	1.52	1.27

As can be seen on Fig. 10, log $\tilde{D}_{AB}$ as defined by equation (7) varies linearly with composition as predicted by equation (4).

REFERENCES

1. L. S. Darken, *Trans. Am. Inst. Min. Engrs* **175**, 184 (1948).
2. J. R. Manning, *Acta Met.* **15,** 817 (1967).
3. S. R. De Groot and P. Mazur, *Non Equilibrium Thermodynamics*. North-Holland (1961).
4. A. Vignes, *Ind. Engng Chem. Fundamentale* **5,** 189 (1966).
5. C. E. Birchenall, *Atoms Movements*, American society for metals, p. 112 (1951).
6. A. D. Le Claire, *Progress in Metal Physics*, edited by B. Chalmers. Butterworths (1949).
7. L. E. Toth and A. W. Searcy, *Trans. Am. Inst. Min. Engrs* **230,** 690 (1964).
8. F. W. Shapink, *Phil. Mag.* **12,** 1055 (1965).
9. D. W. Seith and A. Kottman, *Angew. Chem.* **64,** 379 (1952).
10. R. W. Balluffi and L. L. Seigle, *J. appl. Phys.* **25,** 607 (1954).
11. W. A. Johnson, *Trans. Am. Inst. Min. Engrs* **147,** 331 (1942).
12. W. C. Mallard, A. B. Gardner, R. F. Bass and L. M. Slifkin, *Phys. Rev.* **129,** 617 (1963).
13. R. Hultgren, R. L. Orr, P. D. Anderson and K. K. Kelley, *Selected Values of Thermodynamic Properties of Metals and Alloys*. Wiley (1963).
14. R. O. Meyer and L. M. Slifkin, *Phys. Rev.* **149,** 556 (1966).
15. M. Hansen, *Constitution of Binary Alloys*. McGraw-Hill (1958).
16. L. C. Da Silva and R. F. Mehl, *Trans. Am. Inst. Min. Engrs* **191,** 755 (1951).
17. D. E. Thomas and C. E. Birchenall, *Trans. Am. Inst. Min. Engrs* **192,** 867 (1952).
18. J. Levasseur and J. Philibert, *C.r. hebd. Seanc. Acad. Sci. Paris* **264,** 380 (1967).
19. C. A. Mackliett, *Phys. Rev.* **109** (1964).
20. S. P. Muraka, M. S. Anand and R. P. Agarwala, A.E.E.T./CD/3
21. T. O. Ziebold, *Ph. Thesis M.I.T.* (1965).
22. A. D. Le Claire, *Phil. Mag.* **10,** 641 (1964).
23. A. Vignes and J. P. Haeussler, *Mem. Scient. Revue. metall.* **63,** 1091 (1966).
24. J. I. Goldstein, R. E. Hanneman and R. E. Ogilvie, *Trans. Am. Inst. Min. Engrs* **233,** 812 (1965).
25. M. Badia and A. Vignes, *C.r. hebd. Seanc. Acad. Sci., Paris* **264,** 1528 (1967).
26. B. Fleisher and J. F. Elliott, *The Physical Chemistry of Metallic Solutions and Intermetallic compounds*, Vol. 1, p. 2 F.H.M.S.O. (1959).
27. J. E. Reynolds, B. L. Averbach and M. Cohen, *Acta Met.* **5,** 29 (1957).
28. E. Lifshin, *Second natn. Conf. Electron Microprobe Analysis*, Boston, June 1967.
29. N. L. Peterson, *Phys. Rev.* **132,** 2471 (1963).